An fMRI analysis of the effects of age on working memory performance

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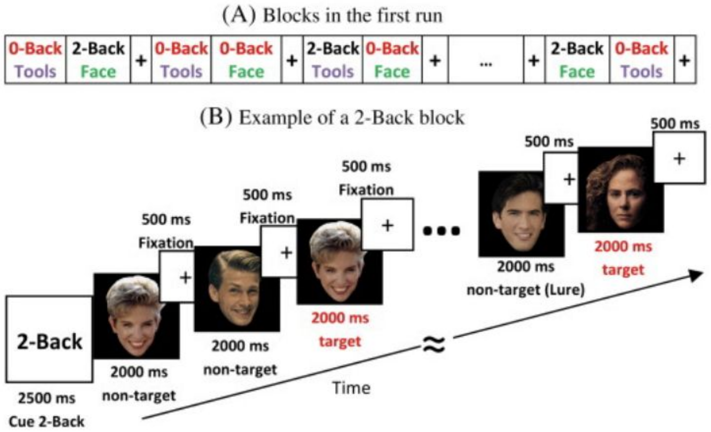
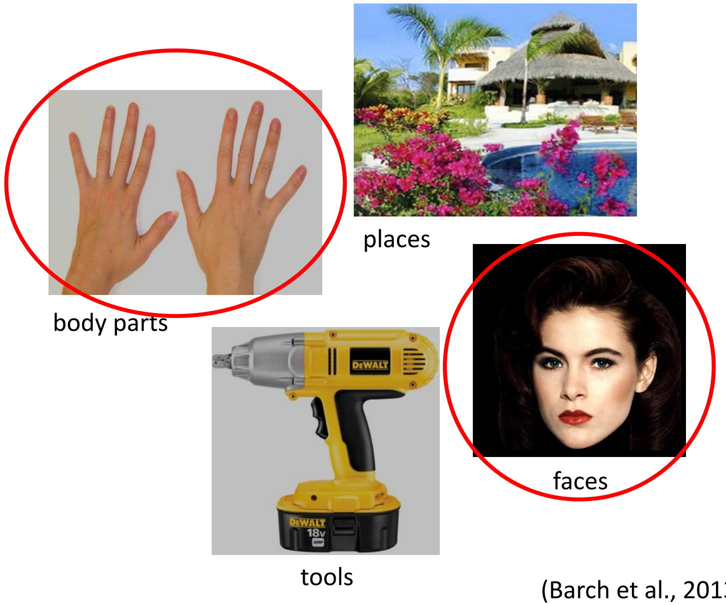
Data Set: Human Connectome Project

- fMRI: data from tasks with behavior
- Subjects performed 2-back working memory task



A. Illustration of the block sequence in the first run of the Working Memory Task. B. Example of the sequence of stimuli in a 2-Back block.

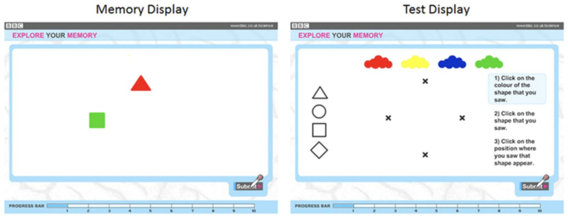
Stimulus set:



(Barch et al., 2013)

Existing Literature:

As we age, working memory deteriorates



Age-related change in visual working memory: a study of 55,753 participants aged 8–75

James R Brockmole [1], Robert H Logie

Affiliations + expand

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Free PMC article

Abstract

Visual working memory (VWM) abilities of 55,753 individuals between the ages of 8 and 75 were assessed to provide the most fine-grain analysis of age-related change in VWM to date. Results showed that VWM changes throughout the lifespan, peaking at age 20. A sharp linear decline follows that is so severe that by age 55, adults possess poorer immediate visual memory than 8 and 9 year olds. These developmental changes were largely explained by changing VWM capacity coupled with small short-term visual feature binding difficulties among children and older adults.

Keywords: ageing; binding; internet; objects; visual working memory.

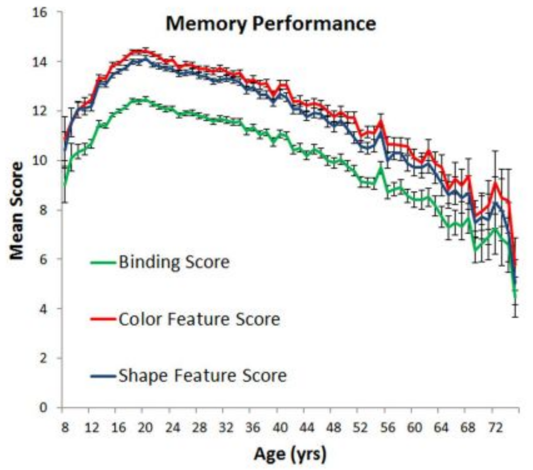


FIGURE 3 | Mean number of objects remembered (with standard error) as a function of age.

(Brockmole & Logie, 2013)

Question of Interest:

How does activation differ across different age groups and stimulus conditions during execution of a 2-back task?

Stimulus Recognition Accuracy (ACC):

ANOVA - WM_Task_2bk_Face_Acc

Cases	Sum of Squares	df	Mean Square	F	p
Age	1314.370	2	657.185	5.894	0.003
Residuals	118862.270	1066	111.503		

Note. Type III Sum of Squares

ANOVA - WM_Task_2bk_Body_Acc

Cases	Sum of Squares	df	Mean Square	F	p
Age	2099.998	2	1049.999	5.116	0.006
Residuals	218786.418	1066	205.241		

Note. Type III Sum of Squares

Stimulus Recognition Response Time (RT):

ANOVA - WM_Task_2bk_Face_Median_RT

Cases	Sum of Squares	df	Mean Square	F	p
Age	100350.839	2	50175.419	1.582	0.206
Residuals	3.381e+7	1066	31720.776		

Note. Type III Sum of Squares

ANOVA - WM_Task_2bk_Body_Median_RT

Cases	Sum of Squares	df	Mean Square	F	p
Age	213541.543	2	106770.772	3.538	0.029
Residuals	3.217e+7	1066	30182.010		

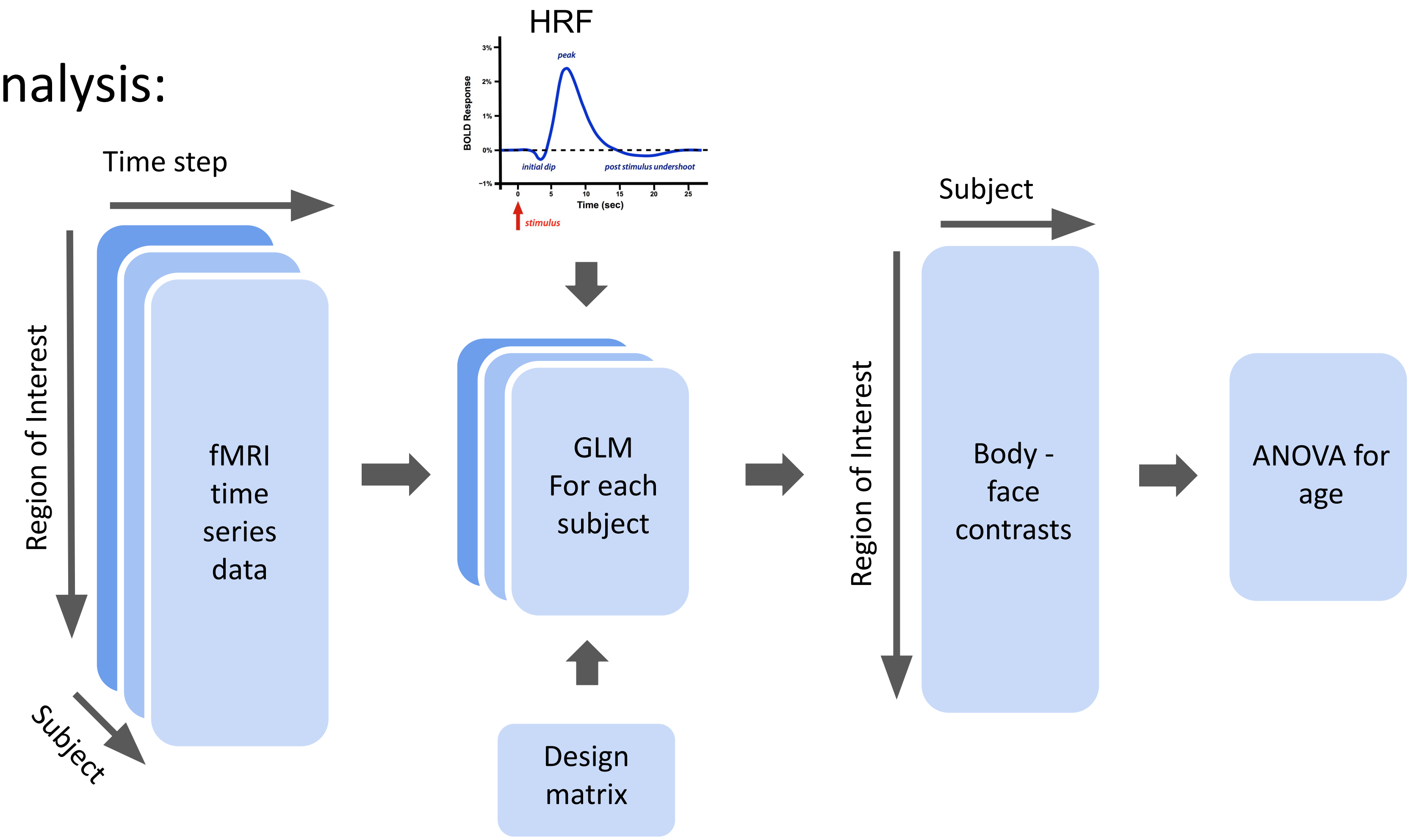
Note. Type III Sum of Squares

Hypothesis:

As subjects *increase in age* while performing the 2-back WM task, they show *slower median response times* and *reduced accuracy*, with this effect being more pronounced when shown body parts.

Therefore, we could quantify and relate activation patterns in certain brain regions to the recognition of both stimuli. We believe these activation patterns will differ between age groups.

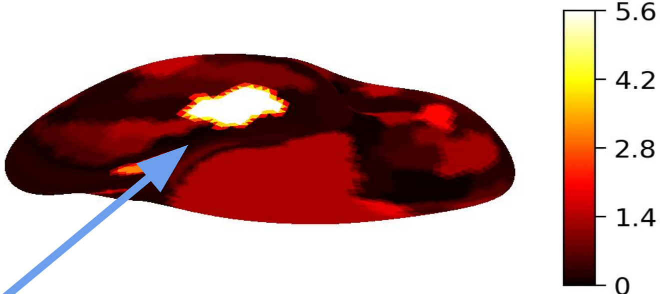
Analysis:


HRF
BOLD Response
3%
2%
1%
0%
−1%
peak
initial dip
post stimulus undershoot
0
5
10
15
20
25
Time (sec)
stimulus
Time step
Region of Interest
Subject
fMRI time series data
GLM For each subject
Design matrix
Subject
Region of Interest
Body - face contrasts
ANOVA for age

Results: Activation Patterns Across Age Groups

Body Parts vs. Faces

Body - Face, Right Hem, ANOVA (Age), Ventral



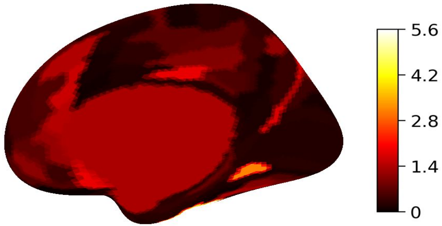
Ventral View

Area TF
$p < 0.005$

Body - Face, Right Hem, ANOVA (Age), Lateral

Lateral View

Body - Face, Right Hem, ANOVA (Age), Medial

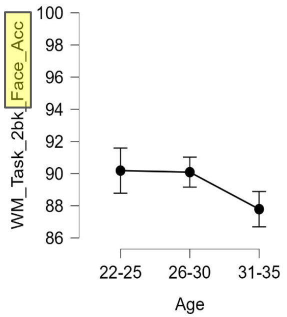


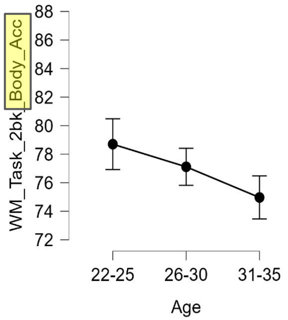
Medial View

Conclusion and future directions:

Based on our results we *proved our initial hypothesis. We were able to find a difference in the activation patterns across age groups in area TF.*

In future studies we hope to analyze the entire HCP dataset, as well as perform a second level GLM.





Stimulus Recognition Accuracy (ACC):

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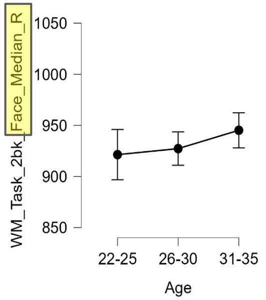

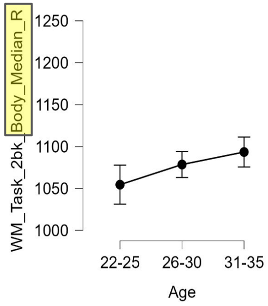

Stimulus Recognition Response Time (RT):

References

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